SOFE 3290 - Software Quality & Project Management

Laboratory 1 - Maven & Unit Testing

Wednesday, Jan 28, 2026

CRN75766

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Introduction

This lab was about learning how to create a Maven project with Java and learn about dependencies, documentation, project structure, and various kinds of unit testing in aJava by creating my own tests for certain functions and tests. The task is/was to build a Maven project of a simple binary calculator and test it and package into a JAR.

Structure

I created a maven project structure and made my own *src/* file that contain *main/* and *test/* directories/files and placed the files form *v1/v2/v3* folders into my own.

Since the binary class only used binary addition, I added OR, AND, & MULTIPLY operations to the BinaryTest.java file. Running *mvn test* let me check if my tests were running correctly, and there were a few failures and issues which I used course material and online resources to help me fix and complete.

mvn site and extended commands from github lab READMEmd; it lets me create an HTML website that has the Javadoc and the test reports. .

Finally, I used *mvn package* to create the executable JAR file with the dependencies included to run the binary calculator.

Conclusion

This lab gave me hands-on experience with Maven, from setting up the project to testing, documenting, and creating a runnable JAR, and let me return to Java basics like classes, methods, and unit tests. Overall, the lab showed how Maven can automate almost every step of building a Java project.

File Edit Selection View Go

SOFE3980U-Lab1

EXPLORER

SOFE3980U-LAB1
- images
- src
  - main \ java \ com \ ontariotechu \ sofe3...
    - App.java U
    - Binary.java U
  - test \ java \ com \ ontariotechu \ sofe39...
    - BinaryTest.java U
- target
- v1
  - App.java 1
  - Binary.java 1
- v2
  - App.java 1
- v3
  - BinaryTest.java 1
- pom.xml
- README.md

OUTLINE
TIMELINE
JAVA PROJECTS
MAVEN

U U | Binary.java ...\sofe3980U U | BinaryTest.java ...\sofe3980U U | App.java v1 1 | Binary.java v1 1 | App.java v2 1 | BinaryTest.java v3 1

src > test > java > com > ontariotechu > sofe3980U > BinaryTest.java > BinaryTest > testMultiplyZero()

```java
public class BinaryTest {
    public void add4() {
    }

    /**
     * Test add with two zeros.
     */
    @Test
    public void add5() {
        Binary binary1 = new Binary("0");
        Binary binary2 = new Binary("0");
        Binary binary3 = Binary.add(binary1, binary2);
        assertTrue(binary3.getValue().equals("0"));
    }

    @Test
    public void testOrSimple() {
        Binary a = new Binary("1010");
        Binary b = new Binary("0011");
        Binary r = Binary.or(a, b);
        assertTrue(r.getValue().equals("1011"));
    }
```

PROBLEMS 6 | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

powershell

```
PS C:\Users\syed5\Documents\SOFE3980U-Lab1> mvn test
[INFO] ------------------------------------------------------------------------
[INFO] BUILD FAILURE
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.503 s
[INFO] Finished at: 2026-03-03T13:51:02-05:00
[INFO] ------------------------------------------------------------------------
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-surefire-plugin:2.22.1:test (default-test) on project BinaryCalculator: There are test failures.
[ERROR]
[ERROR] Please refer to C:\Users\syed5\Documents\SOFE3980U-Lab1\target\surefire-reports for the individual test results.
[ERROR] Please refer to dump files (if any exist) [date].dump, [date]-jvmRun[N].dump and [date].dumpstream.
[ERROR] -> [Help 1]
```

File Edit Selection View Go

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- pom.xml
- README.md

OUTLINE
TIMELINE
JAVA PROJECTS

U U | Binary.java ...\sofe3980U U | BinaryTest.java ...\sofe3980U U | App.java v1 1 | Binary.java v1 1 | App.java v2 1 | BinaryTest.java v3 1

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        Binary binary1 = new Binary("0");
        Binary binary2 = new Binary("0");
        Binary binary3 = Binary.add(binary1, binary2);
        assertTrue(binary3.getValue().equals("0"));
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    public void testOrSimple() {
        Binary a = new Binary("1010");
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        assertTrue(r.getValue().equals("1011"));
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```

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powershell

```
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/release/maven-release/3.0.1/maven-release-3.0.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/release/maven-release/3.0.1/maven-release-3.0.1.pom (11 kB at 229 kB/s)
[INFO] Generating "Plugins" report      --- maven-project-info-reports-plugin:3.0.0:plugins
[INFO] Generating "Summary" report      --- maven-project-info-reports-plugin:3.0.0:summary
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  42.844 s
[INFO] Finished at: 2026-03-03T13:45:48-05:00
[INFO] ------------------------------------------------------------------------
```

```
PS C:\Users\syed5\Documents\SOFE3980U-Lab1> mvn test
[INFO] Results:
[INFO]
[INFO] Tests run: 17, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  6.908 s
[INFO] Finished at: 2026-03-03T14:02:49-05:00
```